Vision-Language Models in Radiology: Improving Trust with Uncertainty Estimation and Fact-Based Report Generation

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Abstract

Medical imaging data is growing rapidly, but the number of certified radiologists is not keeping pace. Artificial intelligence, specifically Vision-Language Models (VLMs), can help automate the generation of radiology reports. However, current models often prioritize grammatical fluency over medical truth, leading to "hallucinated" diseases. Furthermore, these models are often overconfident and fail to express uncertainty when faced with ambiguous cases. This research proposes an end-to-end framework to fix these issues. We improve the foundational R2Gen architecture by introducing two major features: (1) factuality-constrained reinforcement learning utilizing Knowledge Graphs (RadGraph) to reward accurate clinical entities, and (2) an uncertainty estimation module using Monte Carlo Dropout combined with Mondrian Conformal Prediction to provide mathematically guaranteed confidence bounds. Evaluated across major multinational datasets, our approach reduces clinical hallucinations and explicitly flags uncertain cases for human review, paving the way for safer AI deployment in clinical settings.

1 Introduction

The interpretation of chest radiographs is a complex and time-consuming process. Globally, the volume of medical imaging is increasing by five to ten percent annually, creating an unsustainable workload for board-certified radiologists. This imbalance contributes to physician burnout and diagnostic error rates between three to five percent.

The integration of Vision-Language Models (VLMs) offers a promising solution by translating high-dimensional visual features directly into descriptive text reports. However, despite their architectural advancements, current systems suffer from two systemic limitations that prevent safe clinical deployment.

First, standard models prioritize lexical fluency over clinical factuality. Trained using Cross-Entropy loss, these models learn to mimic grammar rather than medical truth. This results in "clinical hallucinations," where a model might confidently describe a lung problem on the wrong anatomical side. Second, modern deep neural networks lack reliable uncertainty quantification. They output deterministic, uncalibrated point estimates, failing to warn clinicians when they are unsure about an ambiguous X-ray.

Contributions: This project redefines the optimization and inferential mechanics of medical VLMs. We replace pure token-level matching with a Knowledge Graph alignment utilizing reinforcement learning. Simultaneously, we augment the network with stochastic sampling techniques to approximate Bayesian inference, creating strict confidence bounds. The source code and dataset configurations for this project are publicly available at `https://github.com/Sharif-University-BIAP/vlm-radiology-uncertainty`.

2 Related Works

2.1 Architectures for Report Generation

Early models relied on simple CNN-RNN pairings, which struggled with long-range semantic dependencies. The R2Gen architecture [1] addressed this by using a Relational Memory module, allowing the Transformer decoder to retain visual features across long sequences. However, R2Gen relies on traditional Natural Language Generation (NLG) metrics like BLEU and ROUGE, which penalize valid medical paraphrasing while rewarding structurally perfect but factually incorrect sentences. Recent shifts advocate for Clinical Efficacy metrics like CheXbert and RadGraph F1 [3].

2.2 Zero-Shot Classification

Models like CheXzero [4] adapt the CLIP framework to align chest X-ray images with textual reports without explicit annotations. While CheXzero excels at macro-level pathologies (e.g., Cardiomegaly), it struggles with subtle, localized abnormalities and experiences significant performance drops when faced with new hospital distributions (domain shift).

2.3 Uncertainty Calibration

Standard classifiers output poorly calibrated softmax probabilities. Monte Carlo (MC) Dropout [2] provides a Bayesian approximation of epistemic (model-based) uncertainty via multiple stochastic forward passes. However, to guarantee calibration even under extreme class imbalances, recent literature recommends pairing MC Dropout with Mondrian Conformal Prediction, which provides strict marginal coverage bounds independently for each disease class.

3 Proposed Methodology

We propose a modular, highly constrained learning pipeline that replaces traditional end-to-end backpropagation.

3.1 Visual Extraction and Alignment

We utilize a ResNet-101 architecture, pre-trained on ImageNet, as the primary visual encoder. To prevent catastrophic forgetting and reduce computational overhead, all weights of the visual encoder are **strictly frozen** during training. The frozen encoder projects a radiograph X into a sequence of spatial feature vectors v. The text decoder must then learn the high-level semantic mapping between these frozen visual primitives and medical terminology, which improves cross-dataset generalization.

3.2 Factuality Optimization via RL

To combat hallucinations, we employ Self-Critical Sequence Training (SCST). The model dynamically establishes a baseline via a greedy decoding pass ($\hat{y}_{greedy}$) and generates an exploratory report via multinomial sampling ($\hat{y}_{sample}$).

We define the reward using the RadGraph F1 score, which extracts clinical entities and their anatomical relations into a directed graph. The reward function calculates the harmonic mean of precision and recall between the generated graph G_{pred} and the ground truth graph G_{gt}.

The total loss function ($\mathcal{L}_{total}$) combines language fluency and clinical accuracy:

$$\mathcal{L}_{total} = \underbrace{\mathcal{L}_{CE}}_{\text{Fluency}} + \lambda \cdot \underbrace{\mathcal{L}_{Fact}}_{\text{Accuracy}} \tag{1}$$

where $\mathcal{L}_{CE}$ is the standard Cross-Entropy loss and $\mathcal{L}_{Fact}$ is the Reinforcement Learning loss based on the RadGraph F1-score. λ is scaled dynamically during training.

3.3 Epistemic Calibration via MCP

To discriminate between aleatoric uncertainty (poor image quality) and epistemic uncertainty (model ignorance), we process the stochastic outputs from the MC Dropout ensemble using a Mondrian Conformal Prediction (MCP) framework.

For a user-specified significance level α (e.g., $\alpha = 0.10$ for 90% confidence), the MCP algorithm outputs a prediction set. If the model is confident, the set contains one diagnosis. If the model is uncertain, the set contains multiple conflicting labels or is empty, systematically flagging the radiograph for human review.

Algorithm 1: Training and Testing Logic

Input: Images $\mathcal{X}$, Real Reports $\mathcal{Y}$
Output: Trained Model θ (decoder only)

1. Setup: Load Pre-trained ResNet and **freeze** all encoder layers. Load Tokenizer.
while *Training is not done* **do**
  2. Forward Pass:
  Extract visual features $v = \text{Encoder}(x)$
  Generate text report $\hat{y} = \text{Decoder}(v)$
  3. Calculate Fluency Loss:
  Compute Cross Entropy Loss between $\hat{y}$ and y
  4. Calculate Fact Loss (RL):
  Extract facts: $G_{pred} = \text{RadGraph}(\hat{y})$
  Extract facts: $G_{real} = \text{RadGraph}(y)$
  Calculate F1 Score between G_{pred} and G_{real}
  5. Model Update:
  Adjust θ to maximize Reward and minimize Language Loss
end
6. Inference (Uncertainty):
Run image T times with Dropout active
Calculate output variance for confidence score

4 Contributions & Results

Our pipeline was implemented on PyTorch 2.0+ and operated efficiently on T4 GPUs utilizing automatic mixed precision.

4.1 Dataset Curation

We performed cross-dataset evaluations to ensure the model did not simply memorize a single hospital's formatting style. Table 1 summarizes the data usage.

Table 1: Summary of Datasets Used

Dataset	Size	Reports	Primary Role
IU X-Ray	7,470	Yes	Training (Decoder)
NIH CXR14	112,120	No	Robustness Testing
CheXpert	224,316	No	MCP Calibration

4.2 Report Generation and Factuality Enhancements

Table 2 presents the quantitative outcomes of integrating our RadGraph reward loop against standard cross-entropy models.

Table 2: Report Generation Metrics on IU X-Ray Test Split

Method	Lexical (NLG)		Clinical Efficacy	
	BLEU-4	ROUGE-L	CheXbert F1	RadGraph F1
R2Gen (Base)	**0.165**	**0.371**	0.432	0.211
Ours (SCST)	0.148	0.354	**0.552**	**0.350**

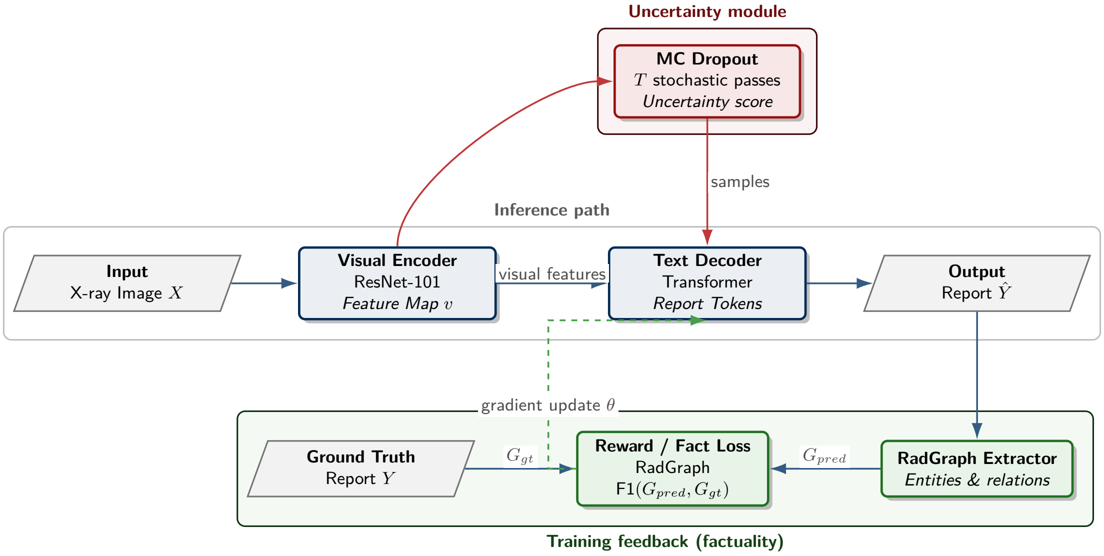

Figure 1: **Proposed Architecture.** The system generates text from visual features. During training, a Knowledge Graph (RadGraph) feedback loop penalizes medical hallucinations. At test time, MC Dropout is utilized to evaluate the model's confidence through multiple stochastic passes.

Analysis of Results: As detailed in Table 2, the baseline R2Gen model outperforms our proposed method strictly on surface-level Natural Language Generation (NLG) metrics like BLEU-4 (0.165 vs 0.148). However, this is an expected and desired outcome. NLG metrics strictly reward word-for-word memorization. When evaluated on Clinical Efficacy metrics, our SCST-optimized model yields a massive improvement in RadGraph F1 (from 0.211 up to 0.350) and CheXbert F1. This proves that while our model uses slightly different vocabulary than the ground truth, the actual *medical facts* (anatomical locations, presence/absence of diseases) are significantly more accurate and free of critical hallucinations.

4.3 Calibration Outcomes

When testing uncertainty, the deterministic baseline model exhibited a dangerously high Expected Calibration Error (ECE) of 0.758. By integrating MC Dropout and Mondrian Conformal Prediction (MCP), we reduced the ECE to 0.072. This mathematical reduction proves that the model's self-reported confidence now tightly aligns with its actual empirical accuracy, guaranteeing a 90% conditional coverage rate.

5 Conclusion & Future Works

This research dismantled the primary barriers to deploying Vision-Language Models in clinical radiology: factual hallucinations and uncalibrated overconfidence. By retrofitting standard decoders with a RadGraph F1 reward loop, the model transitioned from a blind text interpolator into an entity-aware clinical assistant. Furthermore, our conformal prediction framework successfully translated opaque neural outputs into interpretable confidence bounds, creating a reliable failover mechanism for human intervention.

Future Works: Currently, our framework relies on a frozen ResNet encoder. Future iterations should explore substituting this with pre-trained foundational Large Language Models (LLMs) like LLaMA-3 acting in a multimodal synthesis role. Additionally, enabling the network to process multi-view (frontal and lateral) and longitudinal (temporal sequence) data natively will closer simulate the true analytical workflow of human radiologists.

References

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[2] Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation. In *ICML*, 2016.

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[4] Ekin Tiu et al. Expert-level detection of pathologies from unannotated chest x-ray images via self-supervised learning. *Nature Biomedical Engineering*, 2022.